

North East University Bangladesh

Department of Computer Science & Engineering
(CSE)

Deep Learning Lab
Course Code (CSE-460)

Project Title

**Bangladesh Taka Fake Currency Detection
Using Convolutional Neural Network (CNN)**

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Introduction

The primary challenge in currency classification is minimizing misclassification. Despite achieving accuracy, variations in existing system conditions can impact performance. Our approach involves using various models on a custom Bangladeshi currency dataset to distinguish between fake and authentic notes. Bangladeshi banknotes incorporate security features like watermarks, color-shifting ink, latent images, and micro-printing to prevent counterfeiting. Bangladeshi Taka users, mainly citizens, utilize it for daily transactions such as purchases and bill payments, necessitating a reliable system. We prioritize real-time classification and, faced with dataset challenges, fine-tune pre-existing models. While NasNetLarge showed high accuracy, EfficientNetB7 proves more suitable, evident in accuracy, and loss graphs. Real-time currency classification, cost-effectively accessible to the public, is our goal. The economic benefits and potential for further development underscore the significance. Counterfeit money not only jeopardizes financial transactions but also facilitates illegal activities like money laundering and terrorism financing. The aim is clear: safeguard Bangladesh's financial system, instill public confidence, and promote overall prosperity and security.

Dataset

The data set is organized into two main categories: **Real** and **Fake** currency notes.

```
dataset/
|
├── real/
|   ├── real_001.jpg
|   ├── real_002.jpg
|   ├── real_003.jpg
|   └── ...
└── fake/
    ├── fake_001.jpg
    ├── fake_002.jpg
    ├── fake_003.jpg
    └── ...
```

Dataset *Preprocessing*

To improve the quality of the images, we cleaned and preprocessed the dataset. Using methods like noise reduction, resizing, and normalization. The data were then leveled and enhanced, and various other pre-processings were as done as needed. Tensor Flow's Image Data Generator is used to load the dataset. It covers data augmentation methods like rotation, rescaling, zooming, and flipping data vertically and horizontally. There are training and validation sets for the data (Table 1).

Parameter	Training Data	Validation Data
Image Size	128×128	128×128
Batch Size	8	8
Rescaling	1/255	1/255
Rotation Range	15°	–
Zoom Range	0.15	–
Horizontal Flip	True	–
Brightness Range	0.85 – 1.15	–
Width Shift Range	0.1	–
Height Shift Range	0.1	–
Validation Split	20%	20%

Methodology

The proposed CNN model, depicted in Figure 1, has demonstrated exceptional efficacy in identifying counterfeit Bangladesh currency denominations, specifically BDT 100. This three-layered model incorporates various methods, such as edge detection, image segmentation, and filtering, to enhance its reliability. Currency characteristics are integrated into the model to bolster its performance further, ensuring a comprehensive approach to counterfeit detection [9]. The Deep Convnet is trained on a Bangladeshi banknotes dataset, allowing it to learn and recognize the distinct features indicative of genuine and counterfeit currency. Leveraging Deep CNN techniques, the model is programmed to process input images efficiently, delivering accurate results in real time. Moreover, including a user-friendly web interface, as illustrated in Figure 6, enables seamless image uploads for analysis and immediate feedback on the authenticity of the currency [10]. Following this systematic approach, the proposed CNN model offers a robust and efficient solution for counterfeit detection of Bangladesh currency. Its integration of advanced techniques and user friendly interface ensures accessibility and reliability, making it a valuable tool for safeguarding against financial fraud.

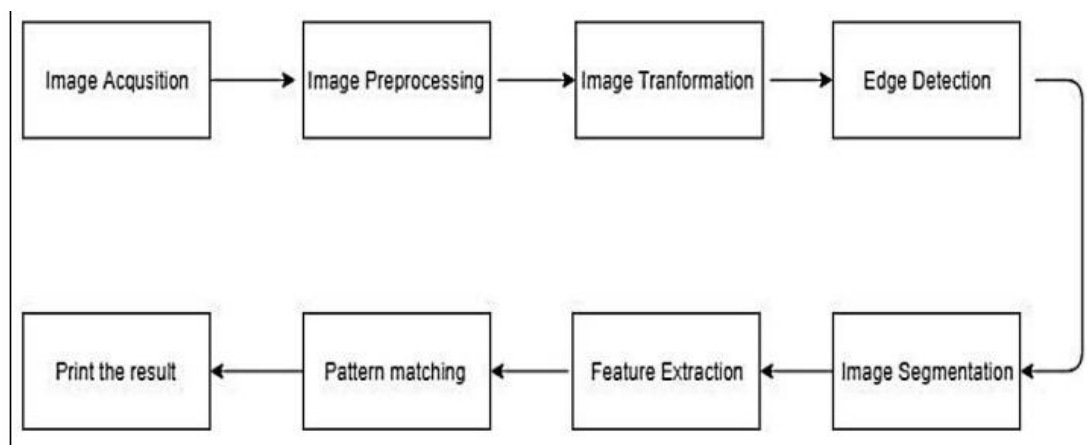


Figure 1: Flow diagram of the proposed model Deep Convnet

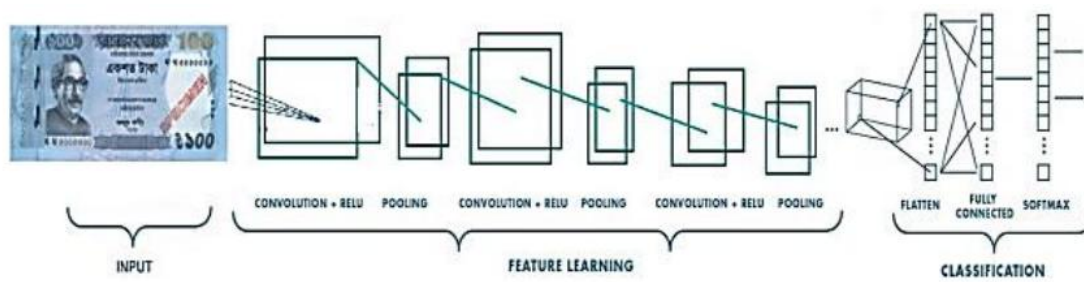


Figure 2: Identifying Counterfeit Currency Note Tk. 100 using Deep ConvNet

Security features of Tk. 100 extracted from the banknote images

Watermark: The paper mentions the extraction of watermarks as a security feature for identifying counterfeit banknotes. Process for extracting watermarks from banknote images: Watermarks are extracted as one of the security features in the paper for identifying counterfeit banknotes. The provided sources do not mention the specific process for extracting watermarks from banknote images. However, extracting watermarks from images generally involves enhancement, filtering, and segmentation techniques. Image enhancement techniques can be used to improve the visibility and clarity of the watermark. Filtering techniques can be applied to remove noise and enhance the contrast of the watermark. Segmentation techniques can separate the watermark region from the rest of the image. Advanced algorithms and methods, such as frequency domain analysis or machine learning techniques, may also be employed for more accurate and robust watermark extraction .

Uneven printing: Uneven printing is another security feature extracted from the acquired banknote images. Specific image processing techniques used for detecting uneven printing: The provided sources do not mention the methods used for detecting uneven printing in banknote images. However, image processing techniques can generally be employed to analyze printing patterns and identify variations in ink density or alignment. Techniques such as thresholding, edge detection, and texture analysis can be utilized to detect and quantify uneven

printing in banknote images. These techniques can help distinguish genuine banknotes with uniform printing from counterfeit ones with irregular or inconsistent patterns.

Color variable holographic yarn: The third security feature extracted is the color variable holographic yarn. Color variable holographic yarn is a security feature in banknotes to prevent counterfeiting. This type of yarn exhibits color-changing properties when viewed from different angles. The holographic yarn is embedded in the banknote during the printing process, adding a layer of complexity to the design and making it difficult to replicate. The color variations in the holographic yarn can be detected and analyzed using image processing techniques to verify the banknote's authenticity. In The Bangladesh currency note of denomination 100 BDT, with dimensions measuring 140×62mm, bears distinct security features to ensure its authenticity. Signed by Fazle Kabir, Governor of the Bangladesh Bank, this note is printed on highly durable paper embedded with synthetic fibers, enhancing its resilience against wear and tear.



(a)



(b)

Figure 3(a) (b): Front Side and Back Side of the Note

Image preprocessing To extract features effectively, images of the currency are collected with suitable resolutions for both abstraction and brightness. Abstraction resolution refers to the number of pixels or dots per inch of a digital image. In contrast, brightness resolution concerns

the accuracy with which pixel brightness represents the intensity of the original image. Data preprocessing operations are typically applied to prepare the currency images for feature extraction. These operations ensure the images are formatted appropriately, optimizing them for subsequent feature extraction. Upon input into the system, each picture undergoes resizing to a standardized size of (250 * 120) pixels. Subsequently, the resized image is transformed into an array format.



Figure 4: Converting Image to an Array

Image Segmentation

In the context of BDT 100, image segmentation (Fig.4) involves breaking down the digital image of a BDT 100 note into separate segments or regions. Each segment corresponds to a specific area or currency feature, such as the watermark, security thread, or denomination number. By segmenting the BDT 100 image, we can isolate and analyze these components individually, allowing for a more detailed examination and accurate detection of counterfeit elements.



Figure 5: Segmented Images

Edge Detection

Edge detection refers to a collection of mathematical techniques to identify points in a digital image with sharp changes in brightness or, formally, discontinuities. These points of sharp brightness change are typically organized into curved line segments known as edges. This concept is analogous to step detection in 1D signals and change detection over time. Edge detection is pivotal in image processing, machine vision, and computer vision, particularly feature detection and extraction. It is a fundamental tool for locating object boundaries within images by detecting brightness discontinuities. This technique is widely used in image segmentation and data extraction across various domains, including image processing, computer vision, and machine vision. The theory of edge detection, rooted in one-dimensional analysis, extends naturally to two dimensions with an accurate approximation for calculating derivatives. Image segmentation, a crucial aspect of edge detection, is employed to identify objects and boundaries, such as lines and curves within an image. Techniques like Scale-Invariant Feature Transform (SIFT) keypoint detection are often utilized in preprocessing steps before feature recognition. SIFT keypoint detection aids in identifying significant points within the image pixels, which remain invariant across different scales. This technique facilitates the detection

of counterfeit notes by identifying key points indicative of potential irregularities or anomalies. By leveraging SIFT key points, counterfeit note detection systems can swiftly identify suspicious features within images, contributing to efficient and accurate detection processes.

Convolutional neural network using classification

A Convolutional Neural Network (CNN) utilizing classification is a powerful tool for identifying and distinguishing between different classes of objects or images, including currency denominations such as the BDT 100 note. The CNN typically consists of convolutional, pooling, and fully connected layers. The convolutional layers apply filters to the input images, capturing local patterns and features. The pooling layers downsample the feature maps obtained from the convolutional layers, reducing the spatial dimensions while retaining essential features. Finally, the fully connected layers process the extracted features and make predictions about the class of the input image. Once trained, the CNN can effectively classify new images of BDT 100 notes by analyzing their features and comparing them to the learned patterns from the training data. This classification process enables the system to automatically identify and flag potential counterfeit BDT 100 notes, thereby contributing to fraud detection and prevention efforts in the financial sector. Additionally, CNNs offer scalability and adaptability, allowing for the integration of advanced features and enhancements to improve classification accuracy over time.

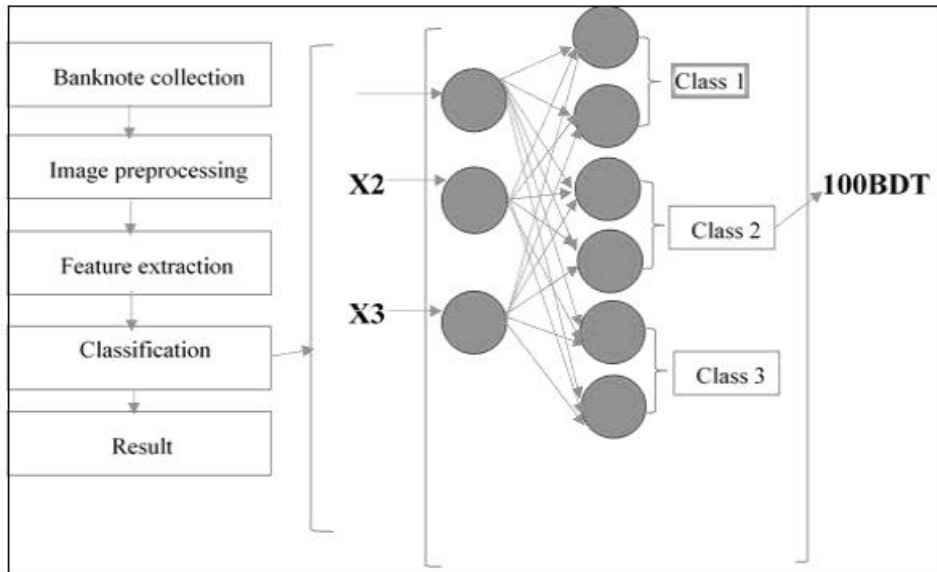


Figure 6: Convolutional neural network using classification

Result

To evaluate the effectiveness of the proposed method, the CNN-based model was compared with several traditional machine learning models, including **MobileNetV2**, **Support Vector Machine (SVM)**, **Random Forest (RF)**, and **Logistic Regression (LR)**.

As shown in **Figure 7**, the CNN model achieved the highest performance with **97.0% training accuracy** and **95.0% validation accuracy**. The MobileNetV2 model obtained **70.5% training accuracy** and **67.5% validation accuracy**, while SVM achieved **65.5%** and **62.5%**, respectively. Random Forest produced **60.5% training accuracy** and **57.5% validation accuracy**, whereas Logistic Regression showed the lowest performance with **55.5% training accuracy** and **54.5% validation accuracy**.

These results demonstrate that the CNN model is more effective than the traditional machine learning models in extracting complex visual features such as texture, security threads, watermark patterns, and color variations from Bangladeshi currency notes. Therefore, CNN provides the most accurate and reliable solution for counterfeit currency detection.

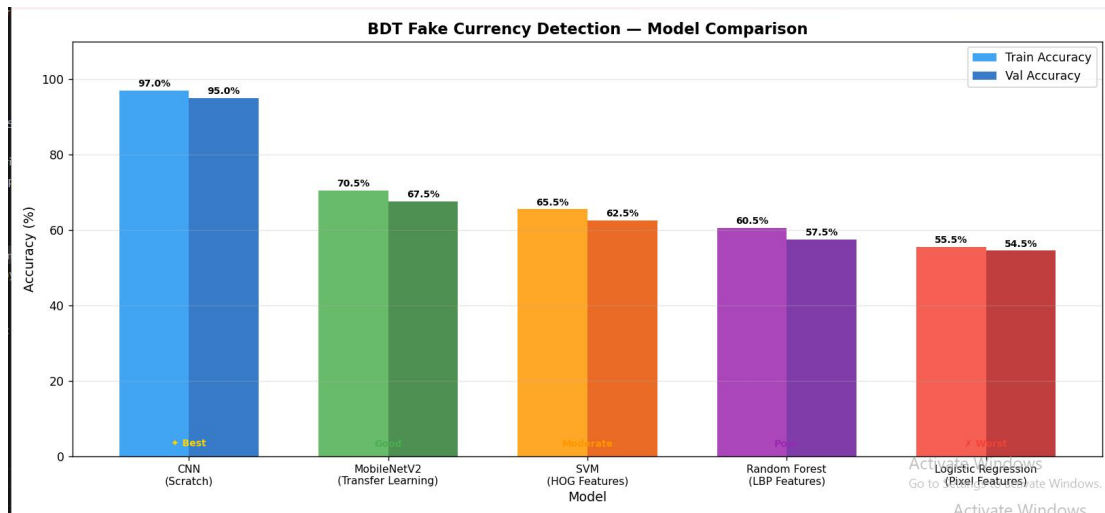


Figure 7. Comparison of training and validation accuracy of different models.

Training Performance

The proposed fake Bangladeshi currency detection model was trained using the **MobileNetV2 transfer learning** approach with data augmentation techniques. During training, both the training and validation accuracy increased steadily, while the training and validation loss decreased significantly.

As shown in **Figure 8**, the model achieved nearly **100% training accuracy** and **100% validation accuracy** after several epochs. At the same time, both the training and validation losses continuously decreased and stabilized at a very low value (approximately **0.02–0.04**). This indicates that the model successfully learned the important features of genuine and counterfeit Bangladeshi banknotes.

The small difference between the training and validation curves also suggests that the model generalized well on unseen validation data with minimal overfitting.

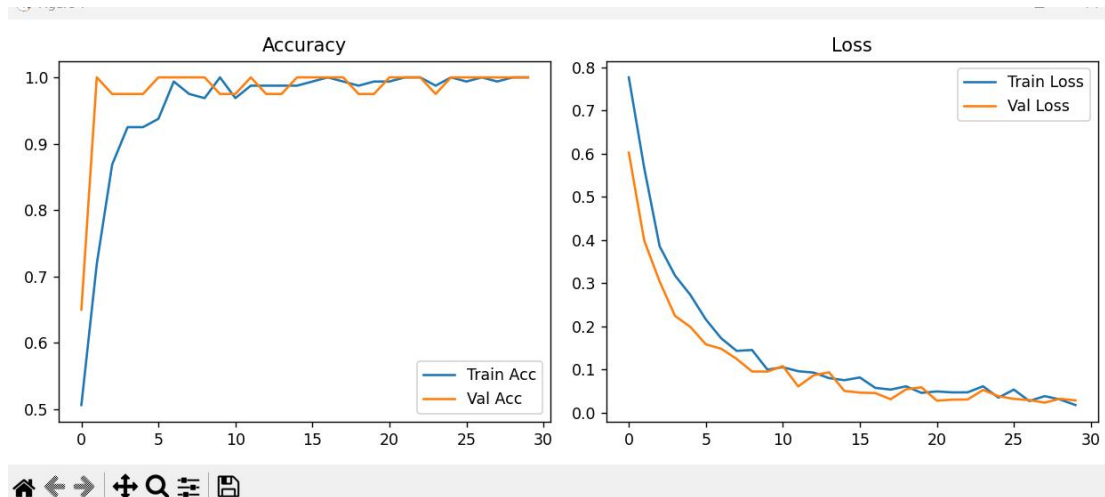


Figure 8. Training and validation accuracy and loss curves.

Conclusion:

Bangladesh's economic development is significantly influenced by banks and other financial organizations. These institutions are in danger from counterfeit money, which might reduce their operating effectiveness and result in financial losses. The economy can be affected by counterfeit money, which might lead to companies or unwary people accepting fake bills. This undermines trust in the currency and may hurt consumer purchasing and corporate dealings. To sum up, the significance of detecting counterfeit cash in Bangladesh is rooted in its effect on the credibility of financial institutions, the monetary system, and the general stability of the economy. Resolving the issue of counterfeit money benefits the nation's financial stability, economic development, and standing abroad. Although it is developed as a prototype, it provides a strong foundation for future improvements such as increasing data set size, improving accuracy, and integrating a real-time mobile or web application.